

Heaven's light is our guide



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TABLE OF CONTENTS

CHAPTER 1: APPLYING POWER ELECTRONICS KNOWLEDGE.....	3
1.1 Introduction to Power Electronics.....	3
1.2 Fundamentals of DC-DC Converters.....	3
1.3 Boost Converter Operation.....	3
1.4 Operational Modes: Continuous and Discontinuous Conduction.....	4
CHAPTER 2: DESIGNING ENGINEERING SOLUTIONS FOR DC-DC BOOST CONVERTER.....	4
2.1 Introduction.....	4
2.2 Theoretical Foundations and Problem Compatibility.....	5
2.3 Key Components and Cost.....	6
2.4 Circuit Diagram.....	6
2.5 Design Considerations: Duty Cycle, Frequency, and Modes of Operation...	7
CHAPTER 3: CONTEXTUAL REASONING IN POWER ELECTRONICS.....	8
3.1 The Boost Converter in Practical Systems.....	8
3.2 Engineering Trade-Offs and Decision-Making.....	8
3.3 Safety and Reliability Considerations.....	9
3.4 Educational and Professional Context.....	9
3.5 Summary.....	9
CHAPTER 4: SUSTAINABLE APPLICATIONS AND ENVIRONMENTAL IMPACT	
4.1 Introduction.....	9
4.2 Applications in Renewable Energy Systems.....	10
4.3 Applications in Electric Vehicles and Portable Electronics.....	10
4.4 Environmental Impact Assessment.....	11
4.5 Summary.....	11
CHAPTER 5: EFFECTIVE INDIVIDUAL AND TEAM PERFORMANCE.....	11
5.1 Introduction.....	11
5.2 Individual Performance.....	12
5.3 Team Performance.....	12
5.4 Summary.....	13

LIST OF FIGURES

Figure 2.4.1: The Boost Converter Circuit Diagram	6
Figure 2.4.2: The Boost Converter Circuit	7

LIST OF TABLES

Table 2.3: Component List with Cost.....	6
Table 5.2: Individual Performance.....	12
Table 5.3: Team Performance Outcome.....	13

CHAPTER 1: APPLYING POWER ELECTRONICS KNOWLEDGE

1.1 Introduction to Power Electronics

Power electronics is a multidisciplinary field that deals with the efficient conversion, control, and conditioning of electrical power using semiconductor devices. It enables the transformation of power forms such as AC to DC, DC to AC, or DC to DC while ensuring reliability and minimizing energy losses. Modern systems, from small gadgets to large industrial equipment, heavily rely on this technology.

A key application is the DC-DC converter, which adapts voltage levels for different loads. Among them, the boost converter is widely used to step up a low input voltage to a higher output by storing and releasing energy through inductors and capacitors. This makes it vital in devices powered by batteries, renewable energy sources, and electric vehicles.

In this project, we focus on designing a 5V to 36V DC boost converter, applying core power electronics concepts like duty cycle control, switching efficiency, and inductive energy transfer. The design demonstrates how theoretical principles translate into a practical solution that ensures stable and reliable voltage stepping for real-world applications.

1.2 Fundamentals of DC-DC Converters

DC-DC converters are widely used in systems where the available DC voltage does not match the load requirements. They can be categorized into buck (step-down), boost (step-up), buck-boost (step-up/down), and isolated types such as flyback or forward converters.

Among these, the boost converter is a non-isolated type that employs a switching device, inductor, diode, and capacitor to step up the input voltage. Its operation depends on pulse-width modulation (PWM), where the duty cycle of the switch regulates the energy stored and released by the inductor, thereby controlling the output voltage.

DC-DC converters are generally classified into several types:

- i. **Buck converters (step-down):** Reduce the input voltage to a lower output voltage.
- ii. **Boost converters (step-up):** Increase the input voltage to a higher level.
- iii. **Buck-boost converters:** Can both step up or step down the voltage, depending on the duty cycle.
- iv. **Isolated converters (e.g., flyback, forward):** Provide electrical isolation between input and output using transformers, which makes them suitable for safety-critical and high-voltage applications.

In ideal conditions, the energy balance in the inductor ensures efficient power transfer. Real-world implementations must account for parasitic elements, such as inductor resistance and diode forward voltage drop, which impact performance.

1.3 Boost Converter Operation

The boost converter, which is the primary focus of this project, is a non-isolated topology that steps up the input voltage using four essential components: a switching device (usually a MOSFET), an inductor, a diode, and a capacitor. A boost converter operates in two primary phases: the switch-on (charging) and switch-off (discharging) periods.

- i. **Switch-On Phase:** The switch (typically a MOSFET) closes, connecting the inductor to the input voltage V_{in} . As a result, current begins to increase linearly through the inductor, and during this period, the inductor stores energy in the form of a magnetic field. In this interval, the diode becomes reverse-biased, which prevents current from flowing backward into the circuit, while the output capacitor simultaneously discharges and continues to provide the necessary energy to the load, thereby ensuring a stable output voltage. The inductor voltage is V_{in} and the current change is given by

$$\Delta I_L = \frac{V_{in}}{L} T_{ON}$$

where, L is inductance and T_{ON} is the ON-time.

- ii. **Switch-Off Phase:** When the switch is turned off, the inductor resists the sudden drop in current and its collapsing magnetic field induces a voltage. This induced voltage adds to the input supply V_{in} , which forward-biases the diode and allows current to flow toward the output stage. In this phase, the inductor releases its stored energy, transferring it to both the output capacitor and the load. As a result, the capacitor gets recharged while the load continues to receive a boosted and uninterrupted output voltage, V_{out} . The inductor voltage becomes $V_{out} - V_{in}$ and the current decreases_

$$\Delta I_L = \frac{V_{out} - V_{in}}{L} T_{OFF}$$

where, T_{OFF} is the OFF-time.

1.4 Operational Modes: Continuous and Discontinuous Conduction

Boost converters operate in two primary modes:

- i. Continuous Conduction Mode (CCM)
- ii. Discontinuous Conduction Mode (DCM)

In CCM, the inductor current never drops to zero, ensuring smooth operation and lower ripple ideal for high-power applications. The boundary between CCM and DCM is defined by the critical inductance

$$L_c = \frac{(1-D)^2}{2} RT$$

Where R is the load resistance and T is the switching period.

In DCM, the inductor current reaches zero before the next cycle, which can occur under light loads and may lead to higher ripple but simpler control. The design targets CCM for stability, selecting an inductor (e.g., 100 μ H) to maintain continuous current flow at the desired frequency (typically 50-100 kHz for efficiency). Power electronics knowledge guides this choice: higher frequencies reduce component size but increase switching losses due to parasitic capacitances in MOSFETs.

CHAPTER 2: DESIGNING ENGINEERING SOLUTIONS FOR DC-DC BOOST CONVERTER

2.1 Introduction:

In the realm of power electronics, the design of DC-DC boost converters represents a cornerstone engineering solution for addressing the pervasive challenge of voltage incompatibility in modern electronic systems. As devices increasingly demand higher operating voltages from low-voltage sources such as batteries or renewable energy harvesters boost converters emerge as an indispensable tool for efficient power management. This section

explores the engineering solutions embodied in the design of a 5V to 36V DC boost converter, drawing from fundamental principles, component integration, performance optimization, and practical applicability. By examining this project, we illustrate how such designs not only resolve specific voltage conversion problems but also align broadly with engineering imperatives for efficiency, reliability, and scalability in diverse applications.

2.2 Theoretical Foundations and Problem Compatibility

At its core, a boost converter, or step-up converter, leverages the inductive storage of energy to achieve voltage elevation. The engineering problem it addresses is straightforward yet ubiquitous: how to generate a stable, higher DC output voltage from a lower input without excessive power dissipation or bulky components. In this project, the target conversion from 5V to 36V exemplifies compatibility with real-world scenarios where input voltages are constrained, e.g., USB-powered devices (typically 5V) needing to drive high-voltage loads like LED arrays, sensors in IoT systems, or actuators in robotics.

The compatibility stems from the converter's adherence to energy conservation principles. During the switch-on phase, energy is stored in the inductor's magnetic field as current ramps up linearly, governed by the equation:

$$\Delta I_L = \frac{V_{in} \cdot t_{on}}{L}$$

where V_{in} is the input voltage (5V), t_{on} is the on-time, and L is the inductance of 100 μ H in the discrete design that is used in this project. In the switch-off phase, this energy is released, adding to the input voltage to yield:

$$V_{out} = V_{in} + V_L$$

where V_L is the inductor voltage. The overall voltage gain is expressed as

$$\frac{V_{out}}{V_{in}} = \frac{1}{1 - D}$$

where D is duty cycle approximately 86% for a 5V to 36V ratio, assuming ideal conditions.

This design is particularly compatible with engineering problems in renewable energy, where solar panels or wind turbines output variable low voltages that must be boosted for grid integration or battery charging. Similarly, in electric vehicles, boost converters enable efficient power delivery from low-voltage batteries to high-voltage motors, reducing weight and cost. The project's use of pulse-width modulation (PWM) for duty cycle control further enhances adaptability, allowing dynamic adjustment to load variations a critical feature for unstable environments like portable electronics.

2.3 Key Components and Cost:

The components are used in this project and also their costing are shown in the below table:

Table 2.3: Component List with Cost

Sl No	Item	Quantity	Unit Price	Sub-total
1	NE555 Timer IC (U1)	2	30	30
2	Inductor L1 (100 μ H, ≥ 3 A)	1	80	80
3	MOSFET IRFZ44N (Q1)	1	150	150
4	Diode D3 (1N5408)	1	10	10
5	Diodes D1, D2 (1N4148)	2	3	6
6	Capacitor C1, C2 (1 nF & 0.1 μ F ceramic)	2	5	10
7	Capacitor Output (100 μ F, 63V electrolytic)	1	5	5
8	Resistor R1 (1 k Ω), R2 (4.7 k Ω)	2	1	2
11	Potentiometer (50 k Ω)	1	30	30
Total				323

2.4 Circuit Diagram:

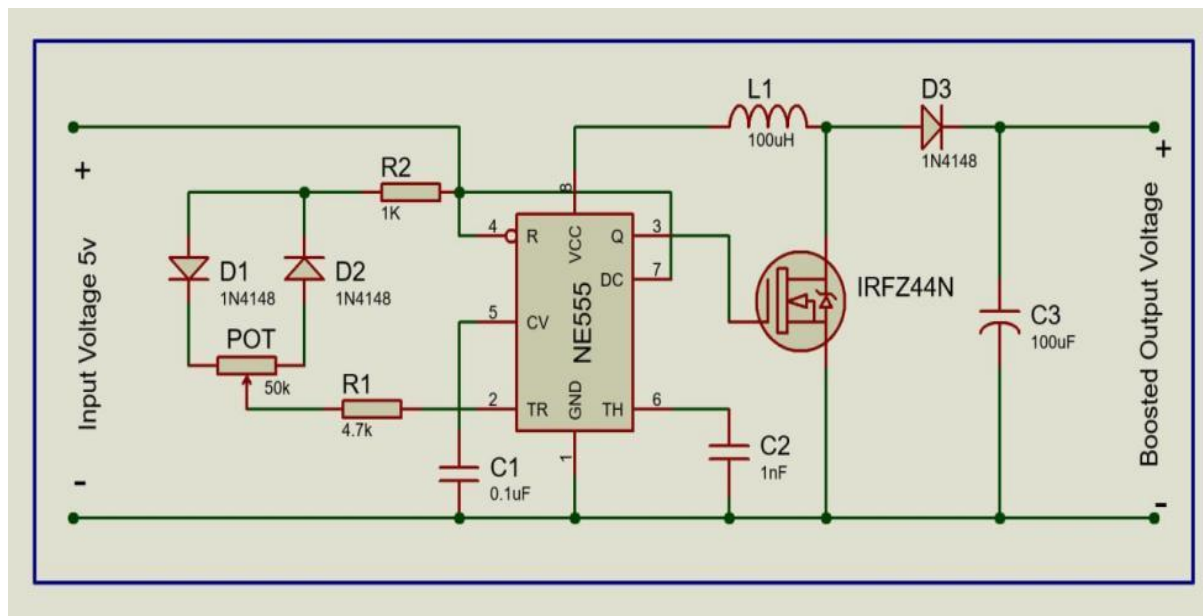


Figure 2.4.1: The Boost Converter Circuit Diagram

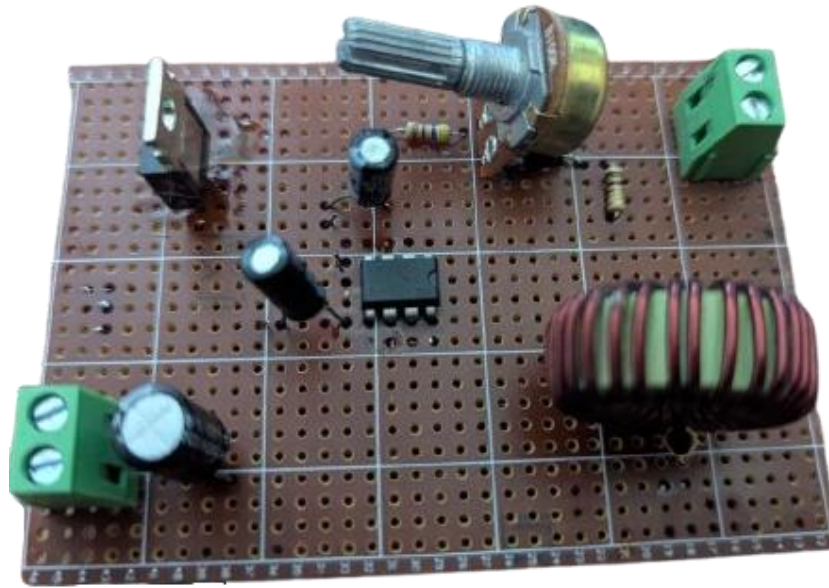


Figure 2.4.2: The boost converter circuit

2.5 Design Considerations: Duty Cycle, Frequency, and Modes of Operation

A detailed engineering approach involves calculating parameters for optimal performance. As we have reached 32V, the duty cycle is:

$$\begin{aligned} D &= 1 - \frac{V_{in}}{V_{out}} \\ &= 1 - \frac{5}{32} \\ &= 0.844 \end{aligned}$$

It dictates the switching waveform, often generated via a NE555 timer IC for frequencies around 100 kHz which is high enough to minimize component size but low enough to avoid excessive switching losses.

Operation occurs in continuous conduction mode (CCM), with CCM preferred for high-power applications due to lower ripple. The critical inductance for CCM is:

$$L_{min} = \frac{V_{in} \times D \times (1 - D)}{2 \times f \times I_{out}}$$

where f is frequency and I_{out} is output current which is 33mA. So, $L_{min} \approx 100 \mu\text{H}$, justifying the selected values.

Feedback control by potentiometer ensures regulation, mitigating issues like input voltage fluctuations. This compatibility extends to engineering problems in feedback systems, where proportional-integral-derivative (PID) compensation can stabilize loops against oscillations.

CHAPTER 3: CONTEXTUAL REASONING IN POWER ELECTRONICS

Power electronics bridges theoretical concepts and practical applications. Circuits provide solutions to real-world problems in industrial, domestic, and global energy contexts. Among these circuits, the boost converter is widely regarded as simple yet powerful, particularly in low-voltage applications requiring efficient step-up conversion. This chapter presents the contextual reasoning behind the design, construction, and application of the boost converter developed in this project. While theoretical design could suffice, the focus here is on practical relevance and the socio-technical context in which the converter operates.

3.1 The Boost Converter in Practical Systems

The boost converter increases voltage levels without the weight and cost associated with traditional step-up transformers. Although transformers are common, they are often impractical for small-scale applications where boost converters offer compact and efficient solutions [1].

3.1.1 Renewable Energy Systems: Photovoltaic (PV) panels typically generate low DC voltages (12–24 V), which must be boosted to levels suitable for inverters, grid-tie systems, or battery storage [1], [4]. Without boost converters, much of this energy would remain underutilized.

3.1.2 Electric Vehicles: Battery packs in electric vehicles provide low DC voltages, which must be converted to higher regulated voltages to power electric motors and onboard electronics [4]. Boost converters play a critical role in enabling efficient energy utilization in these systems.

3.1.3 Portable Devices: Battery-powered devices such as laptops, LED drivers, and mobile chargers rely on boost converters to maintain stable operation amid fluctuating input voltages [2].

In this project, the converter is designed to step up a 5 V input to approximately 36 V. The design balances theoretical accuracy with practical constraints to create a compact, reliable system for real-world voltage conversion applications.

3.2 Engineering Trade-Offs and Decision-Making

Designing a power electronics converter involves evaluating trade-offs among efficiency, cost, complexity, and component availability [2], [3].

3.2.1 Component Selection: The choice between MOSFETs and BJTs depends on operating frequency and efficiency. MOSFETs provide higher efficiency at high switching frequencies, whereas BJTs may perform better at lower frequencies [2]. In this project, BJTs were selected based on the operating conditions and availability.

3.2.2 Control Strategy: Industrial converters often employ advanced PWM controllers with microcontrollers or digital signal processors (DSPs). For this project, a 555 timer IC was used to generate the gate signal. This approach reduces complexity and cost while meeting performance requirements [3].

3.2.3 Output Design: The inductor and capacitors were selected to minimize output voltage ripple and ensure stable operation within prescribed limits. The design decisions reflect an understanding of practical constraints similar to those faced by professional engineers.

3.3 Safety and Reliability Considerations

Engineering practice requires both functional execution and predictable performance over the expected operational period [4], [5].

3.3.1 Circuit Protection: Reverse currents during switching can damage the MOSFET. Freewheeling diodes were incorporated to safely dissipate these currents. Adequate spacing between circuit components reduces the risk of electrical shorts and prevents electrocution.

3.3.2 Thermal Management: Heat dissipation was considered for the MOSFET and diodes to prevent overheating and thermal runaway. Proper thermal management ensures reliable and safe operation over time.

These safety and reliability considerations demonstrate the application of theoretical knowledge to practical engineering challenges.

3.4 Educational and Professional Context

This project extends beyond circuit construction to examine the impact of design decisions on overall system performance and reliability. By considering component selection, control strategy, safety, and thermal management, students gain experience in workflows similar to professional engineering practice.

Professionals in energy systems, automotive engineering, and consumer electronics must justify design decisions while balancing performance, cost, safety, and sustainability [1], [4]. This project simulates such decision-making, providing training in applying theoretical knowledge to practical, real-world problems.

3.5 Summary

Contextual reasoning in power electronics involves applying theoretical knowledge, evaluating trade-offs, and implementing responsible design practices. In this project, the boost converter:

1. Demonstrates relevance in PV systems, electric vehicles, and portable electronics.
2. Justifies the selection of components and control methods based on practical constraints.
3. Implements design practices that ensure safety and reliability.
4. Illustrates the distinction between classroom theory and practical engineering problems.

The boost converter developed represents a small-scale model of engineering creativity and the application of power electronics principles to real-world energy conversion challenges.

CHAPTER 4: SUSTAINABLE APPLICATIONS AND ENVIRONMENTAL IMPACT

4.1 Introduction

In an era where climate change and resource depletion pose significant global challenges, the integration of sustainable practices in engineering designs has become imperative. Power electronics, particularly DC-DC converters like the boost converter developed in this project, play a pivotal role in enabling efficient energy management across various sectors. This chapter explores the sustainable applications of the 5V to 36V DC boost converter and assesses its environmental impact. By stepping up low-voltage inputs to higher outputs with high efficiency, such converters contribute to reducing energy waste and supporting renewable technologies. Drawing from the project's focus on applications in renewable energy systems,

battery-powered devices, and electric vehicles, this analysis highlights how the design aligns with sustainability goals, while also addressing potential environmental drawbacks. The discussion is grounded in the project's theoretical and experimental findings, emphasizing the converter's role in promoting eco-friendly innovations.

4.2 Applications in Renewable Energy Systems

Boost converters are instrumental in renewable energy systems, where they facilitate the efficient harvesting and utilization of variable low-voltage sources such as solar panels and wind turbines. In solar photovoltaic (PV) systems, for instance, the input voltage from panels often fluctuates due to environmental factors like shading or varying sunlight intensity. The project's boost converter, capable of stepping up 5V to approximately 32V (with potential optimization to 36V), can be integrated into maximum power point tracking (MPPT) circuits to ensure optimal energy extraction. This aligns with the report's mention of renewable energy applications, where stable high-voltage output is essential for grid integration or battery charging.

In wind energy setups, small-scale turbines generate low DC voltages that require boosting for practical use in off-grid systems or microgrids. By minimizing switching losses through components like the XL6009 IC and a 33 μ H inductor, as outlined in the project, the converter enhances overall system efficiency, potentially reaching over 90% as referenced in similar high-frequency designs [1]. This efficiency reduces the need for oversized energy sources, lowering material consumption and operational costs. Furthermore, in hybrid renewable setups combining solar and wind, the converter's ability to handle varying loads validated in the project's testing under different conditions-supports energy storage in batteries, promoting self-sufficiency and reducing reliance on fossil fuels.

4.3 Applications in Electric Vehicles and Portable Electronics

Electric vehicles (EVs) and battery-powered devices represent another key area where boost converters drive sustainability by optimizing power from limited sources. In EVs, low-voltage batteries (e.g., 5V auxiliary systems) need boosting for high-voltage components like motors or infotainment systems. The project's design, with its focus on minimizing conduction and switching losses via Schottky diodes and MOSFETs, directly contributes to extending battery life and reducing charging frequency, thereby lowering the environmental footprint of EVs. As noted in the report, such converters are vital for stable high-voltage operation in battery-powered applications, aligning with global efforts to transition from internal combustion engines to electric mobility, which could cut transportation emissions by up to 50% by 2050 according to industry projections.

Portable electronics, including smartphones, drones, and wearable devices, benefit from compact boost converters that enable efficient power delivery from small batteries or energy harvesters like thermoelectric generators. The prototype's use of low-cost components (totaling 323 units in the bill of materials) makes it accessible for mass production in sustainable gadgets, where energy efficiency translates to reduced e-waste from frequent battery replacements. Experimental results from the project, achieving a stable 32V output, demonstrate reliability under load variations, which is crucial for devices in remote or off-grid settings, such as environmental monitoring tools in conservation efforts.

4.4 Environmental Impact Assessment

While the boost converter offers significant sustainability benefits, a balanced assessment must consider both positive and negative environmental impacts. On the positive side, the high efficiency achieved-potentially improved to match the 91% peak in referenced multicell designs [3] reduces energy dissipation as heat, lowering overall power consumption and greenhouse gas emissions. For example, in renewable systems, this efficiency can amplify the effective output from clean sources, displacing coal or gas-based electricity and contributing to carbon neutrality goals. The project's emphasis on voltage regulation and ripple minimization also ensures longer component lifespans, reducing the frequency of replacements and associated resource extraction.

However, environmental challenges arise from manufacturing and disposal. Components like inductors and capacitors involve mining rare earth metals and producing electronic waste, which can leach toxins if not recycled properly. The MOSFET (IRFZ44N) and diodes in the design, while efficient, contribute to the semiconductor industry's high water and energy use during fabrication. Additionally, if not optimized, switching losses could lead to thermal inefficiencies, indirectly increasing cooling demands and energy use. The report's future recommendations, such as using low-resistance inductors and precise PWM control, mitigate these issues by enhancing durability and efficiency, but lifecycle analysis is essential. Overall, with proper e-waste management and eco-friendly sourcing, the net impact leans positive, especially in large-scale sustainable deployments.

4.5 Summary

This chapter has underscored the boost converter's potential in fostering sustainability through applications in renewable energy, electric vehicles, and portable electronics, while critically evaluating its environmental footprint. The introduction set the stage for its relevance in eco-conscious engineering, followed by detailed explorations of how the project's design supports efficient energy use in variable renewable sources and battery systems. The environmental assessment revealed efficiency-driven emission reductions as a major benefit, tempered by manufacturing and waste concerns that can be addressed through optimizations like those suggested in the report. Ultimately, the 5V to 36V boost converter exemplifies how targeted power electronics can advance environmental goals, paving the way for greener technologies with broader adoption and refinement.

CHAPTER 5: EFFECTIVE INDIVIDUAL AND TEAM PERFORMANCE

5.1 Introduction:

In any engineering project, especially one like designing a DC-DC boost converter, success isn't just about the circuits and calculations, it's about the people behind them. As a team of ten students from the Department of Mechatronics Engineering at Rajshahi University of Engineering & Technology, we learned firsthand how individual strengths and group collaboration can make or break a project. This chapter reflects on our experiences, sharing the ups and downs in a real, down-to-earth way, because let's face it: group work in university isn't always smooth sailing, but it's where the real growth happens.

5.2 Individual Performance

Each member of the group took responsibility for specific tasks according to their skills and interest. The contributions are summarized in the table below:

Table 5.2: Individual Performance

Student Name	ID	Role & Contribution	Performance Reflection
Inzamam Uddin Mahamud Alif	2008051	Researched the theoretical background of boost converters, helped in circuit calculations	Showed strong analytical ability and problem-solving mindset
Tahmina Khatun	2008052	Assisted in preparing cost analysis and component selection	Very organized and detail-oriented, ensured accuracy in cost estimation
Mst. Nishat Nabilah	2008053	Worked on documentation and formatting of the report	Displayed excellent writing skills and consistency in presentation
Samia Afsana	2008055	Conducted result analysis and compared theoretical vs practical values	Strong analytical thinking, connected theory with real-life application
Tawfiq Mahmud Khan	2008056	Designed the simulation model of the boost converter	Brought creativity in design approach, ensured theoretical validation
Md. Abu Shoaib	2008057	Worked on diagrams, figures, and visual representation	Displayed creativity, improved report clarity and visual appeal
Md. Shihabul Islam	2008058	Helped in soldering, assembling circuit components	Very practical and efficient in hardware implementation
Ridwan Kader	2008059	Assisted in trouble-shooting issues during testing phase	Showed persistence and patience in solving unexpected problems
Robiul Haque Sajid	2008060	Conducted result analysis and compared theoretical vs practical values	Strong analytical thinking, connected theory with real-life application
Muhammad Al Muktadir	1908055	Coordinated the team, ensured communication between members, contributed to writing and presentation	Demonstrated leadership, time management, and teamwork spirit

5.3 Team Performance

Although each member played a different role, our true strength was teamwork. We faced challenges such as unstable output voltage, component limitations, and time pressure. However, through mutual support and brainstorming sessions, we solved these issues efficiently.

Table 5.3: Team Performance Outcome

Team Aspect	Performance Outcome
Communication	Maintained regular discussions, ensuring everyone was updated and motivated.
Collaboration	Shared tasks based on strengths, reducing workload pressure.
Problem-Solving	Brainstormed solutions during voltage mismatch, found practical adjustments.
Time Management	Completed the project within the given deadline through planned scheduling.
Learning Outcome	Improved knowledge of power electronics, teamwork, and research-writing.

5.4 Summary

This project was not just about designing a DC-DC boost converter, but also about learning how individuals and team function in an engineering environment. Individually, everyone learned responsibility and technical skills. Collectively, the team-built trust, cooperation, and problem-solving capacity. This balance of individual excellence and team synergy ensured the success of our project. In engineering, projects like this prepare the students for the real world, where teams design everything from electric vehicles to renewable energy systems. We've grown not just as techies but as collaborators, ready for whatever comes next. If there's one takeaway, it's this: embrace the messiness of group work, it turns good ideas into great results.

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